Anish Manchanda

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Education

University School of Information Communication & Technology (USICT) — New Delhi, India
B.Tech in Computer Science (Artificial Intelligence), 2024–2028 — *CGPA: 8.75/10*

- **AI & Machine Learning:** Neural Networks, Supervised/Unsupervised Learning, NLP, LLM workflows
- **Data & Systems:** DSA, DBMS (SQL, Normalization), Operating Systems, Computer Networks (TCP/IP)
- **Core CS:** Object-Oriented Programming (Java/Python), Automata Theory, and File Systems

Bal Bharati Public School — New Delhi, India
Senior Secondary (Class 12): 90.8% | *Secondary (Class 10): 91.6%* — *CBSE Board*

Internships

Pepper Labs AI | *AI/ML Intern* — *Jun–Aug 2026*

- **Integrated multiple LLM providers** (OpenAI, Gemini, Groq, SambaNova) behind a unified abstraction layer with **failure handling and provider fallback**, cutting response latency by **40%**.
- **Diagnosed and fixed a triple-invocation concurrency defect** in the request chain, profiling the call path to cut redundant API calls by **66% (3x → 1x)** and hardening retry and error-handling logic.
- **Shipped a production RAG pipeline** to bookbuddy.pepperlabsai.com with hybrid **FAISS + BM25** retrieval, cutting hallucinated responses by **35%**.

Indian Audit & Accounts Department | *Computer Vision Intern* — *Feb–Mar 2026*

- **Developed an OCR data pipeline** in **Python**, digitizing handwritten rural invoices into a structured **SQL** database despite noisy inputs.
- **Built reporting dashboards** with Python and **Chart.js** (time-series graphs, pie charts), giving stakeholders visibility into digitization volume and error rates.
- **Developed and tested** the OCR solution; received a **written appreciation note** (Jul 2026).

Projects

Sorted | *Python, OpenAI/Cerebras LLMs, scikit-learn, SQLite, Telegram, MCP* — GitHub

- **Built an LLM agent system that compiles its own repeated work into tested code**: after 3 similar bills, a compiler agent writes a parser plus tests that must pass **AST safety checks**, a **no-network sandbox** and held-out tests, cutting LLM calls **∼60% (12 → 5/month)** in a 6-month simulation.
- **Designed self-repair and safety**: shadow mode, LLM fallback and **v2 recompile** when a bank changed its email layout, PII redaction, OTP filter, **prompt-injection guard**; parsers scored **100% field accuracy** on 76 synthetic messages.
- **Trained a TF-IDF + logistic-regression classifier** with a confidence threshold (confident predictions **100% accurate**, the rest go to the LLM), plus cost and privacy tracking with token budgets.
- **Shipped a Telegram bot** (reminders, medicine tracker, no-LLM reply rules), an **MCP server** for Claude Desktop and Cursor, and a TinyFish web-agent integration, backed by **54 offline tests**.

MaterialMover | *React, Node.js, FastAPI, PostgreSQL, MongoDB, Docker* — Live | GitHub

- **Built a full-stack marketplace** for construction waste reuse (UN SDG 12): **React/Vite** SPA, **Node.js/Express** API, JWT auth, Cloudinary images, **PostgreSQL** and **MongoDB**.
- **Engineered an agentic AI advisor** with **RAG**: a **FastAPI** service runs hybrid **Sentence Transformer + BM25** search over **200+ records** before each LLM call.
- **Applied prompt engineering** (tool calling, system roles, chain-of-thought) to ground the agent in verified data.
- **Containerized the 4-service architecture** with **Docker** and a **GitHub Actions CI/CD** pipeline, sustaining **99% uptime** for **50+ users**.

Technical Skills

Languages & Databases: Python, TypeScript/JavaScript, SQL, Java, C/C++, Go; PostgreSQL, MongoDB, SQLite
AI & Data Tech: RAG, LLM Integration, AI Agents, MCP, FAISS, BM25, Sentence Transformers, scikit-learn, OCR
Frameworks & Tools: FastAPI, Node.js, Express, React, Chart.js, Docker, GitHub Actions CI/CD, AWS, Git, Linux
Problem Solving: 280+ problems solved across LeetCode, Codeforces and takeUforward (164 on LeetCode)